

Cyril Jacob

Software Engineer | Distributed Systems | Cloud & Infrastructure

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Summary

Software engineering student graduating in 2027 with hands-on experience in backend and distributed systems, cloud infrastructure, DevOps, and CI/CD. Builds production-oriented systems with REST APIs, databases, Docker, Azure, and Linux, supported by AI/ML and computer vision expertise.

Skills

Programming: Python, SQL, Shell Scripting

Backend & Software Engineering: REST APIs, Microservices, CI/CD, Git, GitHub, RAG Schema Design

Databases & Distributed Systems: PostgreSQL, MongoDB, MySQL, Redis, Apache Kafka, MQTT, ChromaDB, HNSW Indexing

Cloud & Infrastructure: Azure, Azure DevOps, Azure Repos, Azure Container Registry, Docker, Kubernetes, Linux, Ubuntu, Netplan, Networking, SSH, Grafana

AI/ML & Computer Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, ONNX, TensorRT, NVIDIA DeepStream SDK, YOLOv8, Mask R-CNN, CLIP, RasterIO

Work Experience

Software Engineering Intern

Jan 2025 – Dec 2025

Graceful Management Systems

- Automated Ubuntu server provisioning with Netplan and shell scripting, reducing deployment time by 70% across production-oriented infrastructure.
- Built software delivery workflows in Azure DevOps using Azure Repos, self-hosted Ubuntu agents, automated testing, CI/CD pipelines, and containerized deployments through Azure Container Registry.
- Deployed containerized microservices to Azure Cloud and designed database schemas and data preparation pipelines for a RAG application and its downstream retrieval and inference workflows.
- Managed Ubuntu servers, virtual machines, permissions, secure SSH access, and dual-interface static IP routing in multi-server network environments.

Computer Vision Engineer (Trainee)

Jul 2024 – Dec 2024

Karunya Innovation and Design Studio

- Deployed real-time CCTV analytics with NVIDIA DeepStream SDK, improving streaming throughput by 40% from 15 to 21 FPS; optimized YOLOv8 and Mask R-CNN with TensorRT INT8 quantisation to reduce latency by 60%.
- Managed Git/GitHub collaboration workflows across a multi-member engineering team building practical computer vision systems.

Projects

Optical-Guided Thermal Super-Resolution

Jan 2025 – Present

Smart India Hackathon 2025 Finalist – ISRO Track

- Developed a physics-guided, self-supervised framework for 2x–4x thermal super-resolution using gated optical–thermal fusion, MTF-based synthetic degradation, and residual reconstruction.
- Applied a Planck/radiance-domain physics constraint; evaluated SSIM, PSNR, and RMSE while mitigating optical texture and hallucination artifacts.

Real-Time Occupancy Analytics System

Sep 2024 – Dec 2024

- Built a real-time system for live campus CCTV streams using Kafka, PostgreSQL, Docker, and Grafana dashboards for floor-wise occupancy insights.

IoT Fleet Management Platform

Oct 2025 – Nov 2025

- Designed a distributed edge orchestration platform for 100+ Raspberry Pi nodes using containerized microservices, PostgreSQL, MongoDB, and Redis.
- Implemented OTA updates with rollback, real-time telemetry, and remote commands through MQTT and REST APIs; monitored device health, network status, and system metrics.

Achievements & Leadership

- Team Lead, Bharatiya Antariksh Hackathon 2025 (ISRO):** Led satellite image super-resolution work to 4th Place Nationally among 61,000+ students across 8,744 teams; achieved 41.45 dB PSNR and 0.97 SSIM on the official ISRO test set.
- Smart India Hackathon 2025:** Grand Finale Top 5 team, ISRO Space Technology Track; quantized and deployed the thermal SR model on NVIDIA Jetson.
- Campus Lead / Organizer, Google Developer Groups On Campus, Karunya (2024–2025):** Led a 25-member student engineering community and coordinated workshops, developer events, and a state-level hackathon.

Education

Karunya Institute of Technology & Sciences

Coimbatore, India | 2023 – Present

B.Tech in Computer Science Engineering (Artificial Intelligence & Machine Learning) | CGPA: 7.75/10 | Graduating 2027

Publication & Certifications

Publications: *Physics-Guided Residual Super-Resolution for Thermal Infrared Satellite Imagery* — Accepted for IEEE ICECA 2026; *Smart Security Management using IoT and HC-05 Bluetooth Module* — IEEE, 2024

Certifications: Microsoft Certified: Azure Fundamentals; SnowPro Associate: Platform Certification; NVIDIA Deep Learning and AI on Jetson Nano; Duke University RAG; University of London Machine Learning for All; Scaler PyTorch; OpenCV Bootcamp